GigaWhat Energy

GigaWhat Gas
Controlled procedure

PR-GAS-045 Leakage Survey of Gas Mains

Field	Value
Document ID	PR-GAS-045
Version	2
Effective from	2025-01-13
Review due	2028-01-13
Owner	Head of Gas Network Operations
Approved by	Gas Safety Manager
Applies to	All GigaWhat Gas sites: low, medium and intermediate pressure mains

FICTIONAL DOCUMENT. Created for the GigaWhat demo. Not for real operational use.

1 Purpose

This procedure sets out how GigaWhat Gas surveys its mains for leakage, how often each main is surveyed, and how leaks found are classified and acted on.

2 Scope

This procedure applies to planned leakage surveys of the LP, MP and IP mains on the GigaWhat gas network. Reported escapes are dealt with under PR-GAS-003, not this procedure.

3 References

- PR-GAS-003 Response to Reported Gas Escapes
- PR-GAS-052 Safe Excavation Near Live Gas Mains
- PR-CORP-004 Lone Working
- PR-CORP-001 Reporting Incidents and Near Misses
- Gas Safety (Management) Regulations 1996 and HSE guidance L80
- Pipelines Safety Regulations 1996

4 Definitions

Surface survey: Walking the line of a main with a sensitive gas detector, sampling air at ground level.

Bar hole test: A test of gas concentration in a bar hole, used to confirm and pinpoint a leak.

ppm: Parts per million of methane in air, as measured by a survey detector.

LEL: Lower explosive limit, about 5% gas-in-air for natural gas.

5 Responsibilities

Head of Gas Network Operations: Sets the annual survey programme and makes sure it is completed.

First Call Operative: Carries out surveys, classifies leaks found and reports them.

Gas Network Controller: Receives category 1 reports immediately and logs all leak reports.

Authorised Person (Gas): Plans repairs of category 2 and 3 leaks and reviews monitoring results.

6 Procedure

6.1 Survey frequencies

Main	Material and pressure	Survey frequency
MAIN-IP-01 GigaWhat to Vale	Steel, IP 7 bar, 300 mm, 1968	Every 12 months
MAIN-MP-03 Vale Road	PE, MP 2 bar, 180 mm, 2009	Every 12 months
MAIN-LP-17 Mill Lane	PE, LP 75 mbar, 125 mm, 2014	Every 3 years
MAIN-LP-22 Station Road	Cast iron, LP 75 mbar, 6 inch, 1932	Every 6 months, plus a frost survey

> NOTE: Carry out a frost survey of MAIN-LP-22 within 48 hours after the ground has been frozen for 3 or more consecutive days. Frozen ground stresses cast iron mains and can cause fractures, while also stopping gas venting through the surface.

6.2 Preparation

6.2.1 Collect the mains plans for the survey route and mark the line of the main.

6.2.2 Check the survey detector (flame ionisation or laser methane) has been calibrated within the last 6 months, and bump test it before starting.

6.2.3 Check the gas-in-air detector for bar hole tests in the same way.

6.2.4 Do not carry out surface surveys when the ground is waterlogged or snow-covered, or in winds above 30 km/h, except frost surveys.

6.2.5 Set up lone working check-ins under PR-CORP-004 and arrange traffic management where the main is in the road.

6.3 Carrying out the survey

6.3.1 Walk the line of the main at no more than 3 km/h, holding the sampling probe within 50 mm of the ground.

6.3.2 Sample at every crack in the surface, kerb line, valve cover, chamber and service entry along the route.

6.3.3 Where the survey detector reads 50 ppm or more, mark the spot with paint and carry out bar hole tests.

6.3.4 Scan with a cable avoidance tool before driving any bar hole, following PR-GAS-052.

6.3.5 Drive bar holes along the line of the main at no more than 2 m intervals around the indication, and test each with the gas-in-air detector.

6.3.6 Record the highest reading, its location and the main ID.

6.4 Leak classification and action

> WARNING: Gas found in or next to a building can reach the flammable range quickly. Do not continue the survey until a category 1 leak has been handed over.

6.4.1 Category 1: gas detected inside any building, or 20% LEL or more in a bar hole within 2 m of a building, or in a chamber or drain. Report it immediately to the Gas Network Controller and stay on site to deal with it as an uncontrolled escape under PR-GAS-003.

6.4.2 Category 2: 20% LEL or more in a bar hole away from buildings, or any confirmed leak on an MP or IP main. Report it to the Gas Network Controller the same day for repair within 5 working days.

6.4.3 Category 3: a confirmed leak on an LP main with bar hole readings below 20% LEL. Report it for repair within 3 months, and re-test monthly until repaired.

6.4.4 Upgrade any category 3 leak to category 2 if re-tests show readings rising.

6.5 Emergency

6.5.1 Call 999 for fire or injury.

6.5.2 If the Gas Network Controller cannot be reached, call the National Gas Emergency Service on 0800 111 999 to report a category 1 leak.

6.5.3 Report any near miss, such as a cable strike while bar holing, under PR-CORP-001.

7 Records

The survey record holds the main ID, date, route, detector serial numbers and calibration dates, weather and ground conditions, and every indication found. Leak reports with their category are logged in the work management system and kept for the life of the main. Completion of the annual survey programme is reported to the Head of Gas Network Operations each quarter.

8 Revision history

Version	Date	Change
1	2021-01-11	First issue.
2	2025-01-13	Survey interval for MAIN-LP-22 reduced from 12 to 6 months and frost survey added. Added category 3 monthly re-tests.